

Japanese SMOG Model

Results and Coefficient Analysis

Model output review

16 September 2026
Saved model results through 2026Q1

1 Results and charts

The historical maximum-likelihood sample is 1995Q1–2022Q4 (112 quarters). The fixed-parameter update extends the state estimates through 2026Q1. At that date the saved smoothed output gap is -0.836% of potential output and the fitted NAIRU is 2.706% . Observed unemployment is 2.673% , about 0.033 percentage points *below* the NAIRU. The negative output gap therefore does not align with a simple current-quarter unemployment-minus-NAIRU reading. The model also contains two lagged unemployment-gap terms, an inflation signal, and a labour-cost signal, so a one-variable interpretation would be misleading.

Smoothed states use the full updated sample. Filtered states use observations available through each plotted quarter while holding the historical coefficients fixed. The plots are model estimates, not direct measurements of potential output or the NAIRU.

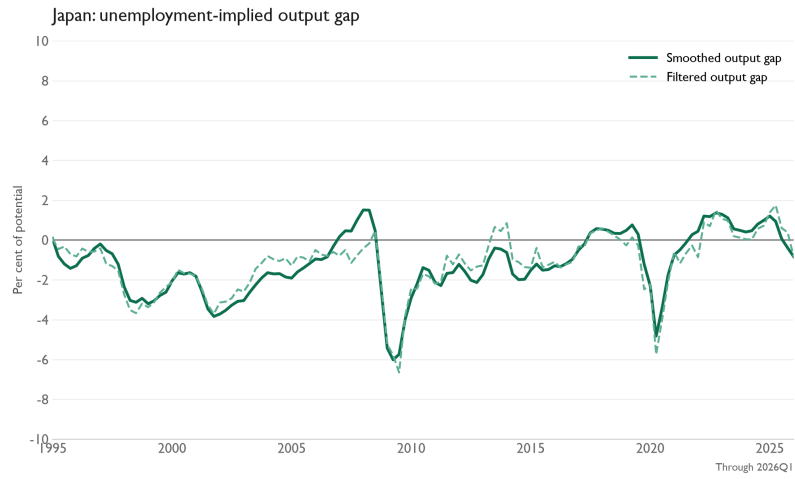


Figure 1: Output gap: smoothed and filtered estimates.

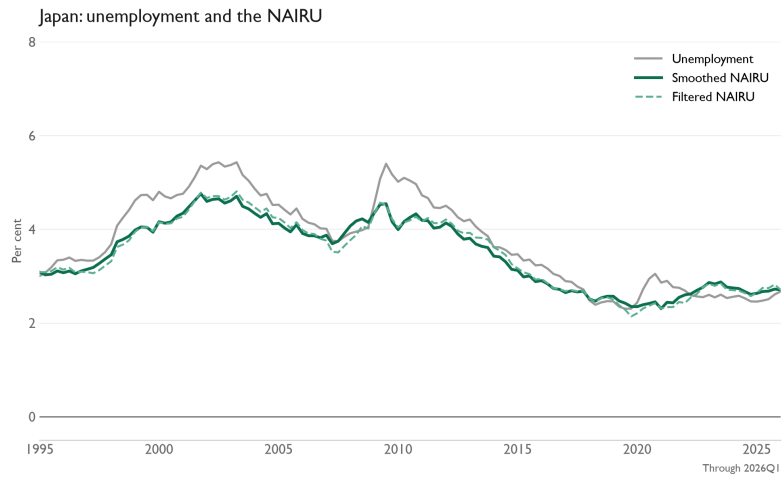


Figure 2: Observed unemployment and smoothed and filtered NAIRU.



Figure 3: NAIRU minus observed unemployment: smoothed and filtered estimates.

2 Reading the current output gap

The model reports a -0.836% output gap in 2026Q1 even though observed unemployment is slightly below its fitted NAIRU. Its two lagged unemployment-gap terms and other signals allow this combination. The latest local data show core inflation of 1.786% and an expectation input of 2.600% . The smoothed gap contributes -0.149 percentage points to the fitted inflation equation. Its full conditional fitted value is 2.560% , so actual inflation is 0.774 points lower. This is a decomposition using the smoothed state, not an out-of-sample forecast.

The negative gap deserves scrutiny, particularly because unemployment is below the fitted NAIRU and the labour-cost equation imposes a large positive loading on unemployment slack. But the observed inflation below the expectation input does not, by itself, show that the gap is too negative or that inflation pressure is being missed. The inflation residual, the unusual labour-cost restriction, and the nearly exact unemployment observation equation all matter for this assessment.

3 State-space structure

Japan's eight states are the log output gap g_t and its two lags, potential log output y_t^* and its lag, and the NAIRU u_t^* and its two lags. The reported output gap is $100g_t$, approximately a percentage deviation from potential. The transition equations are

$$g_t = \phi_1 g_{t-1} + \phi_2 g_{t-2} + \eta_t^g, \quad (1)$$

$$y_t^* = y_{t-1}^* + g^* + \eta_t^y, \quad (2)$$

$$u_t^* = u_{t-1}^* + \eta_t^u. \quad (3)$$

The gap is AR(2), potential output has a quarterly log drift g^* , and the NAIRU is a random walk. The code uses independent Gaussian observation and state shocks with diagonal covariance matrices. Boosted HP trends from the historical data set the known initial state, with zero initial covariance. They also help construct optimizer starting values; Python estimates the coefficients from the Japanese signals.

4 Observation equations

The four measured signals are log GDP y_t , unemployment u_t , core inflation π_t , and annual labour-cost growth n_t . Inflation expectations π_t^e come from the Tankan-labelled input. The import-price term m_{t-1} is a lagged four-quarter level change. Let $D_t = 1$ from 2013Q1 onward and zero before that. The implemented equations are

$$y_t = y_t^* + g_t + \varepsilon_t^y, \quad (4)$$

$$u_t = u_t^* + \lambda_3 g_t + \rho_1(u_{t-1} - u_{t-1}^*) + \rho_2(u_{t-2} - u_{t-2}^*) + \varepsilon_t^u, \quad (5)$$

$$\begin{aligned} \pi_t = & D_t \{ (1 - \beta_1 - \beta_2) \pi_t^e + \beta_1 \pi_{t-1} + \beta_2 \pi_{t-2} \} \\ & + (1 - D_t) \{ (1 - \beta_1 - \beta_2 - \beta_3 - \gamma) \pi_t^e + \beta_1 \pi_{t-1} + \beta_2 \pi_{t-2} + \beta_3 \pi_{t-3} + \gamma n_{t-1} \} \\ & + \lambda_1 g_t + \psi m_{t-1} + \varepsilon_t^\pi, \end{aligned} \quad (6)$$

$$n_t = (1 - \beta_4) \pi_t^e + \beta_4 \pi_{t-1} + L(u_{t-1} - u_{t-1}^*) + \varepsilon_t^n, \quad (7)$$

$$L = 8 + \frac{1}{1 + \exp(-\lambda_2)}. \quad (8)$$

The labour-cost loading is constrained to lie strictly between 8 and 9. It is L , rather than the raw coefficient λ_2 , that multiplies the unemployment gap. A one percentage point output gap corresponds to $g_t = 0.01$ in the inflation and unemployment equations.

5 Fitted coefficients and interpretation

Table 1: Japanese maximum-likelihood coefficients. Values are rounded.

Parameter	Estimate	Meaning in the implemented model
β_1	0.770747	Weight on one-quarter lagged inflation in both regimes.
β_2	-0.091067	Weight on two-quarter lagged inflation in both regimes.
β_3	-0.044532	Weight on three-quarter lagged inflation before 2013Q1.
β_4	2.176609	Weight on lagged inflation in labour-cost growth; the implied expectation weight is -1.1766 .
γ	-0.010815	Loading of lagged labour-cost growth in pre-2013 inflation.
λ_1	17.859318	Inflation loading on the log output gap; a one percentage point output gap adds about 0.1786 percentage points through this term.
λ_2	6.884643	Raw parameter transformed into the positive labour-gap loading $L = 8.99898$; it is not the direct loading itself.
λ_3	-4.011514	Unemployment loading on the log output gap; a one percentage point gap contributes about -0.0401 percentage points.
ρ_1	1.304940	Loading of the first lagged unemployment-minus-NAIRU gap.
ρ_2	-0.482033	Loading of the second lagged unemployment-minus-NAIRU gap.
ϕ_1	1.210914	First AR coefficient of the latent output gap.
ϕ_2	-0.319172	Second AR coefficient of the latent output gap.
ψ	0.002892	Inflation loading of the lagged four-quarter import-price change.
g^*	0.001678	Quarterly potential-log-output drift, about 0.67% at an annualized log-growth rate.
ε_1	0.006487	Standard deviation of log GDP measurement noise, about 0.649% in log-output terms.
ε_2	0.000010	Standard deviation of unemployment measurement noise, in percentage points; at the imposed lower bound.
ε_3	0.285608	Standard deviation of inflation measurement noise, in percentage points.
ε_4	5.349199	Standard deviation of labour-cost-growth measurement noise, in percentage points.
ε_5	0.008176	Standard deviation of the log output-gap state shock, about 0.818% of potential output.
ε_6	0.005007	Standard deviation of the potential-log-output state shock.
ε_7	0.112964	Standard deviation of the quarterly NAIRU state shock, in unemployment percentage points.

The implied inflation-expectation weights are 0.3203 from 2013Q1 and 0.3757 before 2013. The labour equation's $L \approx 8.999$ is a large, positive loading: holding its other terms fixed, one percentage point more unemployment above the previous NAIRU raises predicted labour-cost growth by about nine percentage points. That sign is unusual for a slack measure. It is inherited from the Japanese specification and constrained to the 8–9 interval, so its magnitude should be read as a structural modelling choice, not an unrestricted empirical estimate. The labour-cost signal also has high measurement noise. The near-zero ε_2 makes unemployment almost an exact measurement conditional on the latent states and is another reason to treat the NAIRU and

output gap as specification-sensitive.

6 Likelihood fit and convergence

SciPy’s bounded L-BFGS-B optimizer maximizes the Kalman-filter log likelihood. The accepted-iteration CSV records all 21 parameters and their full-sample likelihood; a second CSV records each quarter’s contribution. The Japanese fit converged after 441 accepted iterations:

Fit statistic	Japan
Historical quarters	112
Initial log likelihood	−137.5153
Iteration 25	23.8740
Iteration 50	24.9822
Final log likelihood	25.7183

The NAIRU innovation scale, $\varepsilon_7 = 0.112964$, is above the imposed data-derived floor of 0.072897. The unemployment observation noise, $\varepsilon_2 = 0.000010$, is at its lower bound. Convergence establishes a numerical optimum from the selected starting point and bounds; it does not establish a global optimum, tight identification, or out-of-sample forecast accuracy. The script does not export standard errors or forecast scores. Quarter-level likelihood terms are Kalman one-step-ahead predictive densities, not direct squared-error measures; early missing regressors cause signals to be masked.

7 Reproduction and source files

From this model folder, run the generation script to re-estimate historical coefficients and overwrite the history CSVs. Then run the update script to apply the fixed fit to later quarters. This report uses the generation code, fitted parameter JSON, both fit-history CSVs, the updated state CSV, the **Update** tab of the local input workbook, and the saved figures. Its numerical tables are a snapshot and do not change automatically when Python is rerun.